Aptitude

1. A newly opened mall

A newly opened shopping mall has five elevators El, E2, E3, E4 and E5. Elevators El, E3 and E5 go to the even floors and elevators E2 and

E4 go to the odd floors. All elevators go to the first and seventh floors.

Which of these choices must be TRUE in the given scenario?

1. A person that enters elevator El on the ground floor cannot reach the fifth floor.
2. A person that enters the elevator E3 on the sixth floor cannot go to the ninth floor.
3. A person must switch elevators to go from an even floor to an odd floor.
4. None of the given options

2. A rectangular plot

A rectangular plot has been alloted by the government for the construction of a embassy building. The length of the plot is increased by 10% to equip a cafeteria.

By what percent should the breadth of the plot be decreased such that the area of the plot is same as before?

1. 120/13
2. 100/11
3. 120/11
4. 100/9

3. Cats and dogs

Given below are 4 statements followed by 4 conclusions numbered 1 to 4. You have to take the given statements to be true even if they seem to be at variance from commonly known facts. Choose the most appropriate options from the given statements, disregarding commonly known facts.

Statements:

Some cats are dogs. No dog is a frog. All frogs are bats. Some bats are eagles.

Conclusions:

1. Some frogs are not eagles.
2. No dog is a bat.

3. Some cats are bats.

4. NO eagle is a cat.

1. Both conclusion 2 & 4 follow

2. Conclusions 2, 3 & 4 follow

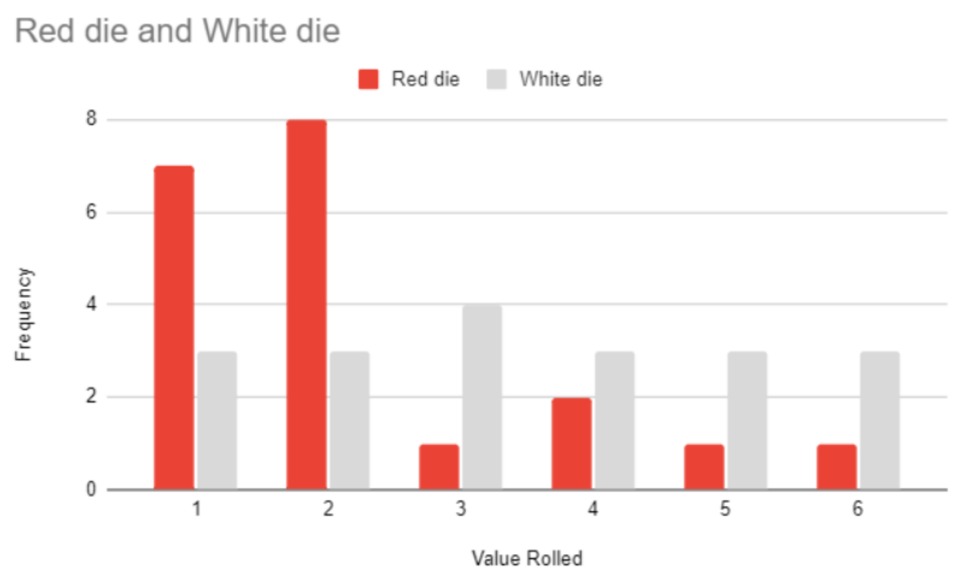
3. Only conclusion I follows

4. No conclusion follows

4. Rolling a die

The given bar graph shows the results from rolling one white and one red dice.

Estimate the arithmetic mean value rolled by the red die.



1. 2.25

2. 3.6

3. 5.8

4. 7.5

5. Capacity of the pool

You are asked to use two pipes P1 and P2 to fill a pool that has another pipe P3 at the bottom of it. It is given that pipes P1 and P2 can fill

the tank in 10 and 20 minutes, respectively and pipe L3 can empty 12 gallons per minute from the pool.

What is the capacity of the pool if all three pipes working together can fill the tank in 30 minutes?

Note: Choose the option that coincides with the nearest integer.

100 gallons

150 gallons

103 gallons

125 gallons

6. Meal prep techniques for busy schedules

A study tested several meal prep techniques T 1, T2, T3, T4 and T5 on how it made lives easier for college students and working individuals

and the following statements were made:

Statements:

1. Tl is twice as effective as T4 but less effective than T5.

2. T 2 is thrice as effective as T 5.

3. T 3 is equally effective as T4.

Which of the following choices are true regarding the following conclusions that are drawn?

Conclusions:

1. T5 has the highest efficacy.

2. T4 and T 5 have the least efficacy.

3. T 2 has the highest efficacy.

Answer Options

Only 1 follows

Only 2 follows

Only 3 follows

Both 1 and 3 follows

7. Row of athletes

In a row of 30 athletes facing North, A1 is 13th from the left end and 6th to the right of A2

What is the position of A2 from the right end of the row?

Answer Options

Select any one option

24 t h

7th

15th

20th

9. Zoey's age

Which of these statements is sufficient to determine the age of Zoey in 2050?

Statement l: When Zoey was born her father was 32 years old, who is 4 years older than Zoey's mother.

Statement 11: Zoey's mother was born in the first leap year that came after 1970.

Answer Options

Select any one option

Statement I ALONE is sufficient, but Statement 11 ALONE is not sufficient. the question

Statement 11 ALONE is sufficient, but Statement I ALONE is not sufficient. the question

BOTH statements TOGETHER are sufficient, but EITHER statement ALONE is insufficient.

EACH statement ALONE is sufficient.

10. Bob's shop

< Previous

Bob is a retailer selling rice in his shop. He mixes two varieties of rice RI and R2 together in the ratio 4:7 and sells this mixture at a price of

Rs. 17 per kg. Due to this, he incurs a loss of 15%.

In the given context, what is the cost price per kg of RI if the cost price of R2 is Rs. 22 per kg?

Answer Options

Rs. 15

Rs. 16.5

Rs. 17

Rs. 17.5

11. Rearranging letters of JUDGEMENT

In how many rearrangements of the word JUDGEMENT, is the letter J positioned in between the 2 Es?

Answer Options

60,480

60,420

50,600

55,600

13. Monica reads

Monica made a schedule for reading books during the five weeks of her literature course. She has borrowed 12 books from her teacher.

The number of pages in each book and the order in which she plans to read the books are shown in the table given below.

She will read exactly 60 pages each day. The only exception will be that she will never begin the next book on the same day that she

finishes the previous one, and therefore on some days, she may read fewer than 60 pages.

At the end of the 32nd day, how many books will Monica have finished?

Book Number	Pages in the book	Total Pages Read
1	260	260
2	120	380
3	115	495
4	170	665
5	165	830
6	60	890
7	200	1090
8	80	1170
9	155	1285
10	95	1380
11	105	1485
12	300	1785

14. The alphanumeric series

Which of these choices comes next in the given alphanumeric series?

KMIO, PR15, UW20, XZ23, ...

Answer Options

CE2

EG22

CE22

EG2

15. Missing value in B

The two grids A and B given below follow a certain pattern. Analyse A and determine the correct value of '?' in B.

$3 \times 5 - 2 \times 6$

35	3	26		57	30	51
45	17	31		62	? 10	21
39	21	32		36	10	42
	A				**B**	

Answer Options

11

10

9

8

16. Dogs and cats

It is given that a dog barks once every two minutes and a cat meows on average once every five minutes.The number of times a dog would bark in 24 hours is approximately equal to what percent greater than the number of times a cat would

meow in that same period?

Answer Options

150%

250%

100%

50%

17. Encoding a message

The message Annual is encoded as ESTKIU, Quarter, is encoded similarly as UZGYJNJ

What is the value of the message Week?

Answer Options

IJKR

IJJR

IKKR

KIJR

19. A chess club

A chess club has three teams T 1, T2 and T3 out of which - one team always tells the truth, one team always lies and one alternates

between telling the truth and lying. The teams get together only thrice a week, i.e., Tuesday, Friday and Sunday. One of these days, the

three teams make the following statements:

Tl: "Tuesday is the most comfortable day to get together and Friday is comfortable for T2".

T2: "Friday is comfortable for T3 and Sunday is comfortable for T 1".

T 3: "Tuesday is the most comfortable day to get together and TI is comfortable with Sunday".

If all teams prefer a different day to get together, then which team is telling the truth?

Note: Al/ teams works as a single union.

Answer Options

T1

T2

T3

Cannot be determined

20. Find ratio of ages

Two years ago, X was six times as old as his son and two years hence, twice his age will be equal to seven times that of his son.

What is the ratio of their present ages?

Answer Options

13:3

12:3

11:4

3:4

Technical

1. Performing an operation

You are given an nxn matrix of integers and asked to perform the following operation on it:

1. Compute the minimum value in each row.

2. Find the maximum of all the minimum values.

What is the time complexity of performing this operation?

Answer Options

O(n)

O(n log n)

O(n^2)

O(n^3)

2. Preventing deadlocks

Read the following statements about deadlocks in an OS and choose the most appropriate option.

A: Permitting resource preemption alone will prevent deadlocks.

B: Preventing circular wait for resources alone will prevent deadlocks.

Answer Options

A and B are both true.

A is true, B is false.

A is false, B is true.

A and B are both false.

4. Using structures

What will the following code snippet print when executed in a 32 bit system?

```
typedef struct {
float angle, radius;
char *next;
```

} VECTOR;

VECTOR v[10];

printf("%d\n", sizeof(v));

Answer Options

120

160

200

320

5. The sign magnitude representation

Which of these choices is the most appropriate sign magnitude representation for the number -15?

Answer Options

1111

1111

11111

1110

6. The Auxiliary Space complexity

You are using the following function to traverse through a circular linked list.

void printList(struct Node *first)

struct Node *temp = first;

if (first NULL)

do

printf("%d temp»data);

temp = temp»next;

while (temp != first);

What would the be Auxiliary Space complexity of the code snippet?

Answer Options

0(1)

O(n)

O(2n)

None of these

7. Invoking the print statement

How many times is the statement print(i,j) invoked in the following code snippet?

for (i=l; i n-1 ; i++)

for (j=i+l; j n; j++)

print(i,j)

Answer Options

Select any one option

n^2 times

n^2 - n times

(n^2 - n)/2 times

n^2/2 times

8. Memory with fixed partitions

A memory consists of fixed size partitions as given in the diagram. There are three process each of which requires 200 units for execution. The processes are allocated memory using the first fit algorithm with contiguous memory allocation policy.

Find the ratio of internal fragmentation to external fragmentation.

600	200	100	500

Answer options

1:1

2:3

2:1

1:2

9. Using 8085 to calculate factorial of a number

You are using the following program to calculate the factorial of a number using a 8085 microprocessor.

LXI H, 2000H

MOV B, M

MVI D, OIH

FACTORIAL: CALL MULTIPLY

DCR B

UNZ FACTORIAL

INX H

MOV M, D

HLT

MULTIPLY: MOV E, B

MVI A, OOH

MULTIPLYLOOP: ADD D

xxx

MOV D, A

RET

What should you use in place of XXX to complete the program?

Answer Options

DCR E

JNZ MULTIPLYLOOP

DCR E

JNZ MULTIPLYLOOP

HLT

IO. Result of executing the code

What will be the result of executing the following code snippet?

float WHATISlT(float x, int n) {

if (n >= 1)

return ((n

else

return ((n

0)? 1 : n-1));

0)? 1 : n+l));

int main(void) {

float tz WHATISlT(2, 3);

float k: WHATISlT(O.5, -3);

printf("%f\n",t);

printf("%f\n",k);

Answer Options

6, 1.5

4, .25

8,8

The program will enter an endless loop.